Fiscal Year:	FY 2018	**Task Last Updated:**	FY 02/20/2018
PI Name:	Blakely, Eleanor A Ph.D.		
Project Title:	Simulation of GCR-Induced Harderian Gland and Lung Tumorigenesis		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cancer**:Risk of Radiation Carcinogenesis		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2014-15 HERO NNJ14ZSA001N-RADIATION. Appendix D: Ground-Based Studies in Space Radiobiology
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No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	7	**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mao, Jian-Hua Ph.D. (Lawrence Berkeley National Laboratory) Chang, Polly Ph.D. (SRI International)		
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Task Description:	Increased cancer risk remains one of the primary concerns for travel into deep space and may preclude manned missions to Mars due to the large uncertainties that currently exist in estimating cancer risk from the spectrum of radiation types found in space with available human epidemiological data on radiation-induced cancer. Existing data on human risk of cancer from exposures to X- and gamma-rays must be scaled to the many types and fluxes of radiations found in space using radiation quality factors and dose-rate modification factors. Linearity of dose-response is often assumed since the shape of the dose-response at doses below 100 mSv is unknown. However, preliminary analysis of our recently completed experiments with silicon and titanium ion suggest non-linear structure in the low-dose region for both Harderian and lung tumorigenesis. The goal of this proposal is to reduce uncertainties in the estimation of particle radiation carcinogenesis from galactic cosmic radiations (GCR). Our approach is to use the existing data on Harderian gland tumorigenesis from investigations of individual high-atomic number, and high-energy (HZE) ion beams to predict outcome of mixed ion chronic GCR exposures, and then measure Harderian gland and lung dose-dependent tumor data for such exposures. We will compare some of the measured cancer prevalence outcomes of a GCR-simulated exposure with the theoretical predictions based on experiments with individual ion beams. Our main hypothesis is that HZE carcinogenic effects are LET- (linear energy transfer) and dose-dependent and not additive at low particle doses due to the contribution of non-targeted effects. Completion of the proposed GCR studies using a mouse strain that has already been well characterized with a low spontaneous tumor background will allow a significant test of modeling approaches used to estimate carcinogenesis risk from a simulated GCR beamline.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Completion of our research on predicting and measuring HZE tumorigenic effects to test the hypothesis that HZE carcinogenic effects and LET- and dose-dependent are not additive at low particle doses due to the contribution of non-targeted effects will allow a significant test of modeling approaches used to estimate carcinogenic risk from single-ion and mixed radiation fields. This work will also benefit estimation of risk of tumorigenesis from mixed radiation fields on Earth.
Task Progress:	*Recent experimental milestones: 1--Participated in NASA Space Radiation Laboratory (NSRL) 17C—Exposed a total (with unirradiated controls) of 171 female CB6F1 female mice to 350 MeV/amu Oxygen ions 0, 20, 40, or 80 cGy. 2--Animal husbandry continuing at Lawrence Berkeley National Laboratory (LBNL) for a total of nearly 400 animals from the single or two-ion beams from NSRL 16C and the animals from the Oxygen run during NSRL 17C run. 3--We have provided murine cardiac tissues collected from our previous Harderian Gland tumorigenesis studies that are currently listed in the NASA Tissue Archive to Dr. David Goukassian and continue to collaborate with him to conduct his NASA-funded research. 4--Murine Bone Marrow Harvest from 52 mice was conducted at 17-weeks post NSRL 16C irradiation for chromosome aberration study. However, the quality of the samples was not optimal and aberrations were not scorable, possibly due to the age and strain of the animals. The assay method was refined by optimizing the stimulating/blocking reagents and incubation conditions using the same strain of aged mice. 5--In collaboration with Dr. Megumi Hada, chromosome aberrations in human lymphocytes ex vivo, induced by single and mixed beam (H + Si) from NSRL 16C were recorded. The dose response of simple chromosome exchanges appears to be linear for H and Si, but the slope of the dose-dependent increase differs by a factor of ~6X. The H dose-response curve appears to be linear-quadratic. Slope of complex exchanges is steeper for Si than H by a factor of ~16X. Silicon-induced total exchanges appear to be linear with dose, proton exchanges appear to be linear quadratic with dose. Chromosome aberrations with mixed beams - consisting of different doses of H followed within 2 min by different doses of Si indicated that the chromosome aberration yield of simple exchanges in mixed beams in sometimes more than additive. The frequency and type of chromosome aberrations induced is dependent on the percentage of the dose from the proton versus the silicon beam. This biological system can detect 10 cGy differences in total dose. 6--In Silico Modeling of murine Harderian Gland tumorigenesis by Prof. Ray Sachs found substantially improved dose-effect relations for 1-ion beams, and confirmed: (a) our conjecture that 95% confidence intervals for baseline no-synergy (+ or -) are substantially tighter when correlations between calibrated model parameters are properly taken into account, (b) the well-known inapplicability of simple effect additivity as a baseline definition of no-synergy (+ or -) for a mixed radiation field when component Dose Effect Ratios (DERs) are highly curvilinear, e.g., are all highly concave at very low doses due to NTE. In addition, Prof. Sachs found a way to generate the mixture baseline DER of incremental effect that is now applicable to a “mixture of mixtures” theorem. This model is relevant because nominally 1-ion beams are usually themselves mixtures at the target, due, e.g., to mouse tissue fragmentation and self-shielding. He has also extended his synergy theory to one special case of constant chronic dose-rate lasting a year or more. Theoretical modeling of chromosome aberration work has progressed and a new paper was published online in December 2017 for print publication in the New Year. 7--Future plans: Necropsy and bone marrow harvest of 226 NSRL 16C mice (February 2018), and 171 NSRL 17A mice (August 2018). We also plan to irradiate a 6-ion mixed beam rapid sequence study in May 2018. *Presentations: 1--Heidelberg Ion Beam Therapy Center, Heidelberg, Germany, Blakely, E., “Charged Particle Therapy”, 21 June 2017. 2--NASA Space Radiation Summer School, Brookhaven National Laboratory, Upton, New York, Blakely, E. “Space Radiation and Cataracts” on 22 June 2017. 3--NASA Space Radiation Summer School, Brookhaven National Laboratory, Upton, New York, Blakely, E. “Charged Particle Therapy” on 22 June 2017.
Bibliography Type:	Description: (Last Updated: 02/12/2026)

Abstracts for Journals and Proceedings	Blakely EA, Mao J-H, Bakke J, Grover A, Rusek A, La Tessa C, Snyder D, Bjornstad KA, Rosen CJ, Sachs RK, Chang P. "Simulating GCR-Induced Tumor Risk." 63rd Annual Meeting of the Radiation Research Society, Grand Fiesta Americana Coral Beach, Cancun, Mexico, October 14-18, 2017. Radiation Research Society 2017 Meeting Abstracts eList. 63rd Annual Meeting of the Radiation Research Society, Grand Fiesta Americana Coral Beach, Cancun, Mexico, October 14-18, 2017. , Oct-2017
Abstracts for Journals and Proceedings	Hada M, Rhone JR, Beitman AJ, Chang PY, Blakely EA, Sachs RK, Saganti PB. "Chromosome Aberrations Induced by Proton and Silicon Mixed Exposure to Human Lymphocytes, Fibroblasts, and Epithelial Cell." 63rd Annual Meeting of the Radiation Research Society, Grand Fiesta Americana Coral Beach, Cancun, Mexico, October 14-18, 2017. Radiation Research Society 2017 Meeting Abstracts eList. 63rd Annual Meeting of the Radiation Research Society, Grand Fiesta Americana Coral Beach, Cancun, Mexico, October 14-18, 2017. , Oct-2017
Abstracts for Journals and Proceedings	Blakely EA. "Open Research Questions in Particle Radiobiology." The Second Shanghai International Summit on Particle Radiation Therapy (SI-SPRT), Shanghai, China, December 8-9, 2017. The Second Shanghai International Summit on Particle Radiation Therapy 2017 Abstract eList. Second Shanghai International Summit on Particle Radiation Therapy (SI-SPRT), Shanghai, China, December 8-9, 2017. , Dec-2017
Abstracts for Journals and Proceedings	Chang P, Hada M, Mao J-H, Bakke J, Grover A, La Tessa C, Synder D, Bjornstad KA, Rosen CJ, Rhone JR, Beitman AJ, Saganti PB, Sachs RK, Blakely EA. "Simulating GCR-Induced Tumor Risk & Chromosome Aberration Correlates." 2018 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2018. 2018 NASA Human Research Program Investigators' Workshop, Galveston, TX, January 22-25, 2018. , Jan-2018
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Awards	Blakely EA. "Distinguished Service Award for her contributions to the Radiation Research Society, presented at the 2017 Annual Meeting of the Radiation Research Society, Cancun, Mexico, October 14-18, 2017." Oct-2017